14 Entropy

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In this chapter we will use the results from Chapter 13 to define a quantity called entropy and to understand how entropy changes in reversible and irreversible processes. We will also consider the statistical basis for entropy, and use this to understand the entropy of mixing, the apparent conundrum of Maxwell's demon and the connection between entropy and probability.

14.1 Definition of entropy

In this section, we introduce a thermodynamic definition of entropy. We begin by recalling from eqn 13.26 that $\oint \mathrm{đ}Q_{\mathrm{rev}}/T = 0$. This means that the integral

$$\int_{\mathrm{A}}^{\mathrm{B}} \frac{\mathrm{đ}Q_{\mathrm{rev}}}{T}$$

is path independent (see Appendix C.7). Therefore the quantity $\mathrm{đ}Q_{\mathrm{rev}}/T$ is an exact differential and we can write down a new state function which we call entropy. We therefore *define* the **entropy** S by

$$\boxed{\mathrm{d}S = \frac{\mathrm{đ}Q_{\mathrm{rev}}}{T},} \tag{14.1}$$

so that

$$S(\mathrm{B}) - S(\mathrm{A}) = \int_{\mathrm{A}}^{\mathrm{B}} \frac{\mathrm{đ}Q_{\mathrm{rev}}}{T}, \tag{14.2}$$

and S is a function of state. For an adiabatic process (a reversible adiathermal process) we have that

$$\mathrm{d}Q_{\mathrm{rev}} = 0. \tag{14.3}$$

Hence an adiabatic process involves no change in entropy (the process is also called **isentropic**).

14.2 Irreversible change

Entropy S is defined in terms of reversible changes of heat. Since S is a state function, then the integral of S around a closed loop is zero, so that

$$\oint \frac{\mathrm{đ}Q_{\mathrm{rev}}}{T} = 0. \tag{14.4}$$

Let us now consider a loop which contains an irreversible section (A→B) and a reversible section (B→A), as shown in Fig. 14.1. The Clausius inequality (eqn 13.34) implies that, integrating around this loop, we have that

$$\oint \frac{đQ}{T} \leq 0. \tag{14.5}$$

Writing out the left-hand side in detail, we have that

$$\int_{\mathrm{A}}^{\mathrm{B}} \frac{đQ}{T} + \int_{\mathrm{B}}^{\mathrm{A}} \frac{đQ_{\mathrm{rev}}}{T} \leq 0, \tag{14.6}$$

and hence rearranging gives

$$\int_{\mathrm{A}}^{\mathrm{B}} \frac{đQ}{T} \leq \int_{\mathrm{A}}^{\mathrm{B}} \frac{đQ_{\mathrm{rev}}}{T}. \tag{14.7}$$

This is true however close A and B get to each other, so in general we can write that the change in entropy dS is given by

$$\mathrm{d}S = \frac{đQ_{\mathrm{rev}}}{T} \geq \frac{đQ}{T}. \tag{14.8}$$

The equality in this expression is only obtained (somewhat trivially) if the process on the right-hand side is actually reversible. Note that because S is a state function, the entropy change in going from A to B is independent of the route.

Consider a thermally isolated system. In such a system $đQ = 0$ for any process, so that the above inequality becomes

$$\boxed{\mathrm{d}S \geq 0.} \tag{14.9}$$

This is a very important equation and is, in fact, another statement of the second law of thermodynamics. It shows that any change for this thermally isolated system always results in the entropy either staying the same (for a reversible change)[1] or increasing (for an irreversible change). This gives us yet another statement of the second law, namely that: 'the entropy of an isolated system tends to a maximum.' We can tentatively apply these ideas to the Universe as a whole, under the assumption that the Universe itself is a thermally isolated system:

Application to the Universe:
Assuming that the Universe can be treated as an isolated system, the first two laws of thermodynamics become:

(1) U_{Universe} = constant.

(2) S_{Universe} can only increase.

The following example illustrates how the entropy of a particular system and a reservoir, as well as the Universe (taken to be the system plus reservoir), changes in an irreversible process.

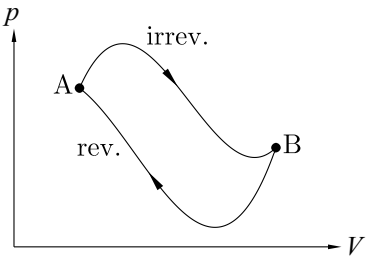


Fig. 14.1 An irreversible and a reversible change between two points A and B in p–V parameter space.

[1] For a reversible process in a thermally isolated system, $T\,\mathrm{d}S \equiv \mathrm{d}Q_{\mathrm{rev}} = 0$ because no heat can flow in or out.

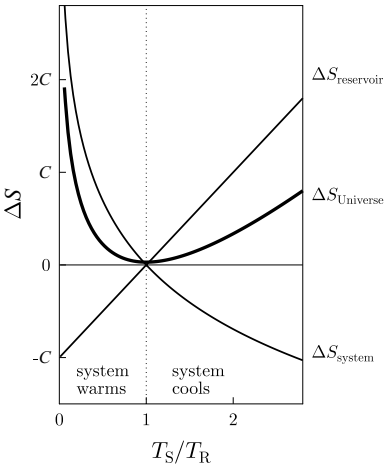


Fig. 14.2 The entropy change in the simple process in which a small system is placed in contact with a large reservoir.

Example 14.1

A large reservoir at temperature $T_{\rm R}$ is placed in thermal contact with a small system at temperature $T_{\rm S}$. They both end up at the temperature of the reservoir, $T_{\rm R}$. The heat transferred from the reservoir to the system is $\Delta Q = C(T_{\rm R} - T_{\rm S})$, where C is the heat capacity of the system.

- If $T_{\rm R} > T_{\rm S}$, heat is transferred from reservoir to system, the system warms and its entropy increases; the entropy of the reservoir decreases, because heat flows out of it.
- If $T_{\rm R} < T_{\rm S}$, heat is transferred from system to reservoir, the system cools and its entropy decreases; the entropy of the reservoir increases, because heat flows into it.

Let us calculate these entropy changes in detail: The entropy change in the reservoir, which has constant temperature $T_{\rm R}$, is

$$\Delta S_{\rm reservoir} = \int \frac{đQ}{T_{\rm R}} = \frac{1}{T_{\rm R}} \int đQ = \frac{\Delta Q}{T_{\rm R}} = \frac{C(T_{\rm S} - T_{\rm R})}{T_{\rm R}}, \qquad (14.10)$$

while the entropy change in the system is

$$\Delta S_{\rm system} = \int \frac{đQ}{T} = \int_{T_{\rm S}}^{T_{\rm R}} \frac{C\,{\rm d}T}{T} = C \ln \frac{T_{\rm R}}{T_{\rm S}}. \qquad (14.11)$$

Hence, the total entropy change in the Universe is

$$\Delta S_{\rm Universe} = \Delta S_{\rm system} + \Delta S_{\rm reservoir} = C \left[\ln \frac{T_{\rm R}}{T_{\rm S}} + \frac{T_{\rm S}}{T_{\rm R}} - 1 \right]. \qquad (14.12)$$

These expressions are plotted in Fig. 14.2 and demonstrate that even though $\Delta S_{\rm reservoir}$ and $\Delta S_{\rm system}$ can each be positive or negative, we always have that

$$\Delta S_{\rm Universe} \geq 0. \qquad (14.13)$$

14.3 The first law revisited

Using our new notion of entropy, it is possible to obtain a much more elegant and useful statement of the first law of thermodynamics. We recall from eqn 11.7 that the first law is given by

$${\rm d}U = đQ + đW. \qquad (14.14)$$

Now, for a reversible change only, we have that

$$đQ = T{\rm d}S \qquad (14.15)$$

and

$$đW = -p\mathrm{d}V. \tag{14.16}$$

Combining these, we find that

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V. \tag{14.17}$$

Constructing this equation, we stress, has assumed that the change is reversible. However, since all the quantities in eqn 14.17 are functions of state, and are therefore path independent, this equation holds for irreversible processes as well! For an irreversible change, $đQ \leq T\,\mathrm{d}S$ and also $đW \geq -p\,\mathrm{d}V$, but with $đQ$ being smaller than for the reversible case and $đW$ being larger than for the reversible case so that $\mathrm{d}U$ is the same whether the change is reversible or irreversible.

Therefore, we *always* have that:

$$\boxed{\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V.} \tag{14.18}$$

This equation implies that the internal energy U changes when either S or V changes. Thus, the function U can be written in terms of the variables S and V which are its so-called **natural variables**. These variables are both *extensive* (i.e. they scale with the size of the system).[2] The variables p and T are both *intensive* (i.e. they do not scale with the size of the system) and behave a bit like *forces*, since they show how the internal energy changes with respect to some parameter. In fact, since mathematically we can write $\mathrm{d}U$ as

[2] See Section 11.1.2.

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_V \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_S \mathrm{d}V, \tag{14.19}$$

we can make the identification of T and p using

$$T = \left(\frac{\partial U}{\partial S}\right)_V \text{ and} \tag{14.20}$$

$$p = -\left(\frac{\partial U}{\partial V}\right)_S. \tag{14.21}$$

The ratio of p and T can also be written in terms of the variables U, S and V, as follows:

$$\frac{p}{T} = -\left(\frac{\partial U}{\partial V}\right)_S \left(\frac{\partial S}{\partial U}\right)_V, \tag{14.22}$$

using the reciprocal theorem (see eqn C.41). Hence

$$\frac{p}{T} = \left(\frac{\partial S}{\partial V}\right)_U, \tag{14.23}$$

using the reciprocity theorem (see eqn C.42). These equations are used in the following example.

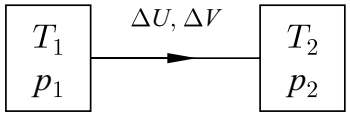


Fig. 14.3 Two systems, 1 and 2, which are able to exchange volume and internal energy.

Example 14.2

Consider two systems, with pressures p_1 and p_2 and temperatures T_1 and T_2. If internal energy ΔU is transferred from system 1 to system 2, and volume ΔV is transferred from system 1 to system 2 (see Fig. 14.3), find the change of entropy. Show that equilibrium results when $T_1 = T_2$ and $p_1 = p_2$.

Solution:

Equation 14.18 can be rewritten as

$$\mathrm{d}S = \frac{1}{T}\mathrm{d}U + \frac{p}{T}\mathrm{d}V. \tag{14.24}$$

If we now apply this to our problem, the change in entropy is then straightforwardly

$$\Delta S = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\Delta U + \left(\frac{p_1}{T_1} - \frac{p_2}{T_2}\right)\Delta V. \tag{14.25}$$

Equation 14.9 shows that the entropy always increases in any physical process. Thus, when equilibrium is achieved, the entropy will have achieved a maximimum, so that $\Delta S = 0$. This means that the joint system cannot increase its entropy by further exchanging volume or internal energy between system 1 and system 2. $\Delta S = 0$ can only be achieved when $T_1 = T_2$ and $p_1 = p_2$.

Eqn 14.18 is an important equation that will be used a great deal in subsequent chapters. Before proceeding, we pause to summarize the most important equations in this section and state their applicability.

Summary

$\mathrm{d}U = đQ + đW$	always true
$đQ = T\,\mathrm{d}S$	only true for reversible changes
$đW = -p\,\mathrm{d}V$	only true for reversible changes
$\mathrm{d}U = T\mathrm{d}S - p\,\mathrm{d}V$	always true

For irreversible changes: $đQ \leq T\,\mathrm{d}S$, $đW \geq -p\,\mathrm{d}V$

14.4 The Joule expansion

In this section, we describe in detail an irreversible process which is known as the **Joule expansion**. One mole of ideal gas (pressure p_i, temperature T_i) is confined to the left-hand side of a thermally isolated container and occupies a volume V_0. The right-hand side of the container (also volume V_0) is evacuated. The tap between the two parts of the container is then suddenly opened and the gas fills the entire container of volume $2V_0$ (and has new temperature T_f and pressure p_f). Both

containers are assumed to be thermally isolated from their surroundings. For the initial state, the ideal gas law implies that

$$p_{\mathrm{i}}V_0 = RT_{\mathrm{i}}, \tag{14.26}$$

and for the final state that

$$p_{\mathrm{f}}(2V_0) = RT_{\mathrm{f}}. \tag{14.27}$$

Since the system is thermally isolated from its surroundings, $\Delta U = 0$. Also, since U is only a function of T for an ideal gas, $\Delta T = 0$ and hence $T_{\mathrm{i}} = T_{\mathrm{f}}$. This implies that $p_{\mathrm{i}}V_0 = p_{\mathrm{f}}(2V_0)$, so that the pressure halves, i.e.

$$p_{\mathrm{f}} = \frac{p_{\mathrm{i}}}{2}. \tag{14.28}$$

It is hard to calculate directly the change of entropy of a gas in a Joule expansion along the route that it takes from its initial state to the final state. The pressure and volume of the system are undefined during the process immediately after the partition is removed since the gas is in a non-equilibrium state. However, entropy is a function of state and therefore for the purposes of the calculation, we can take another route from the initial state to the final state since changes of functions of state are independent of the route taken. Let us calculate the change in entropy for a reversible isothermal expansion of the gas from volume V_0 to volume $2V_0$ (as indicated in Fig. 14.5). Since the internal energy is constant in the isothermal expansion of an ideal gas, $\mathrm{d}U = 0$, and hence the new form of the first law in eqn 14.18 gives us $T\,\mathrm{d}S = p\,\mathrm{d}V$, so that

$$\Delta S = \int_{\mathrm{i}}^{\mathrm{f}} \mathrm{d}S = \int_{V_0}^{2V_0} \frac{p\,\mathrm{d}V}{T} = \int_{V_0}^{2V_0} \frac{R\,\mathrm{d}V}{V} = R\ln 2. \tag{14.29}$$

Since S is a function of state, this increase in entropy $R\ln 2$ is also the change of entropy for the Joule expansion.

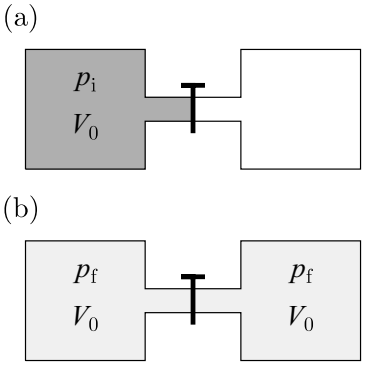


Fig. 14.4 The Joule expansion between volume V_0 and volume $2V_0$. One mole of ideal gas (pressure p_{i}, temperature T_{i}) is confined to the left-hand side of a container in a volume V_0. The container is thermally isolated from its surroundings. The tap between the two parts of the container is then suddenly opened and the gas fills the entire container of volume $2V_0$ (and has new temperature T_{f} and pressure p_{f}).

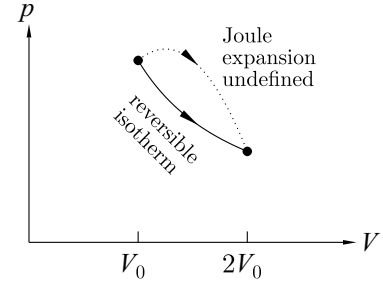


Fig. 14.5 The Joule expansion between volume V_0 and volume $2V_0$ and a reversible isothermal expansion of a gas between the same volumes. The path in the p–V plane for the Joule expansion is undefined, whereas it is well defined for the reversible isothermal expansion. In each case however, the start and end points are well defined. Since entropy is a function of state, the change in entropy for the two processes is the same, regardless of route.

Example 14.3

What is the change of entropy in the gas, surroundings and Universe during a Joule expansion?

Solution:

Above, we have worked out ΔS_{gas} for the reversible isothermal expansion and the Joule expansion: they have to be the same. What about the surroundings and the Universe in each case?

For the reversible isothermal expansion of the gas, we deduce the change of entropy in the surroundings so that the entropy in the Universe does not increase (because we are dealing with a reversible situation).

$$\begin{aligned} \Delta S_{\mathrm{gas}} &= R\ln 2, \\ \Delta S_{\mathrm{surroundings}} &= -R\ln 2, \\ \Delta S_{\mathrm{Universe}} &= \Delta S_{\mathrm{gas}} + \Delta S_{\mathrm{surroundings}} = 0. \end{aligned} \tag{14.30}$$

Notice that the entropy of the surroundings goes down. This does not contradict the second law of thermodynamics. The entropy of something can decrease if that something is not isolated. Here the surroundings are not isolated because they are able to exchange heat with the system.

For the Joule expansion, the system is thermally isolated so that the entropy of the surroundings does not change. Hence

$$\begin{aligned} \Delta S_{\text{gas}} &= R\ln 2, \\ \Delta S_{\text{surroundings}} &= 0, \\ \Delta S_{\text{Universe}} &= \Delta S_{\text{gas}} + \Delta S_{\text{surroundings}} = R\ln 2. \end{aligned} \tag{14.31}$$

Once the Joule expansion has occurred, you can only put the gas back in the left-hand side by compressing it. The best[3] you can do is to do this reversibly, by a reversible isothermal compression, which takes work ΔW given (for 1 mole of gas) by

$$\Delta W = -\int_{2V_0}^{V_0} p\,\mathrm{d}V = -\int_{2V_0}^{V_0} \frac{RT}{V}\,\mathrm{d}V = RT\ln 2 = T\Delta S_{\text{gas}}. \tag{14.32}$$

The increase of entropy in a Joule expansion is thus $\Delta W/T$.

[3] In other words, the method involving the least work.

A paradox?:

- In the Joule expansion, the system is thermally isolated so no heat can be exchanged: $\Delta Q = 0$.
- No work is done: $\Delta W = 0$.
- Hence $\Delta U = 0$ (so for an ideal gas, $\Delta T = 0$).
- But if $\Delta Q = 0$, doesn't that imply that $\Delta S = \Delta Q/T = 0$?

The above reasoning is correct, until the very end: the answer to the question in the last point is NO! The equation $đQ = T\mathrm{d}S$ is only true for reversible changes. In general $đQ \le T\mathrm{d}S$, and here we have $\Delta Q = 0$ and $\Delta S = R\ln 2$, so we have that $\Delta Q \le T\Delta S$.

14.5 The statistical basis for entropy

We now want to show that as well as defining entropy via thermodynamics, i.e. using $\mathrm{d}S = đQ_{\text{rev}}/T$, it is also possible to define entropy via statistics. We will motivate this as follows:

As we showed in eqn 14.20, the first law $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$ implies that

$$T = \left(\frac{\partial U}{\partial S}\right)_V, \tag{14.33}$$

or equivalently

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_V . \tag{14.34}$$

Now, recall from eqn 4.7 that

$$\frac{1}{k_\mathrm{B}T} = \frac{\mathrm{d}\ln\Omega}{\mathrm{d}E}. \tag{14.35}$$

Comparing these last two equations motivates the identification of S with $k_\mathrm{B}\ln\Omega$, i.e.

$$\boxed{S = k_\mathrm{B}\ln\Omega.} \tag{14.36}$$

This is the expression for the entropy of a system which is in a particular macrostate in terms of Ω, the number of microstates associated with that macrostate. We are assuming that the system is in a particular macrostate which has fixed energy, and this situation is known as the *microcanonical ensemble* (see Section 4.5). Later in this chapter (see Section 14.8), and also later in the book, we will generalize this result to express the entropy for more complicated situations. Nevertheless, this expression is sufficiently important that it was inscribed on Boltzmann's tombstone, although on the tombstone the symbol Ω is written as a 'W'.[4] In the following example, we will apply this expression to understanding the Joule expansion which we introduced in Section 14.4.

[4] See page 29.

Example 14.4

Joule expansion:
Following a Joule expansion, each molecule can be either on the left-hand side or the right-hand side of the container. For each molecule there are therefore two ways of placing it. For one mole (N_A molecules) there are 2^{N_A} ways of placing them. The additional number of microstates associated with the gas being in a container twice as big as the initial volume is therefore given by

$$\Omega = 2^{N_\mathrm{A}} \tag{14.37}$$

for one mole (N_A molecules) of gas, so that

$$\Delta S = k_\mathrm{B}\ln 2^{N_\mathrm{A}} = k_\mathrm{B}N_\mathrm{A}\ln 2 = R\ln 2, \tag{14.38}$$

which is the same expression as written in eqn 14.29.

14.6 The entropy of mixing

Consider two different ideal gases (call them 1 and 2) which are in separate vessels with volumes xV and $(1-x)V$ respectively at the same

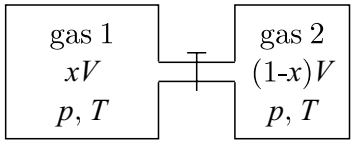


Fig. 14.6 Gas 1 is confined in a vessel of volume xV, while gas 2 is confined in a vessel of volume $(1-x)V$. Both gases are at pressure p and temperature T. Mixing occurs once the tap on the pipe connecting the two vessels is opened.

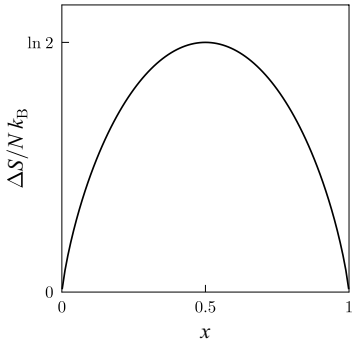


Fig. 14.7 The entropy of mixing according to eqn 14.40.

pressures p and temperatures T (see Fig. 14.6). Since the pressures and temperatures are the same on each side, and since $p = (N/V)k_{\mathrm{B}}T$, the number of molecules of gas 1 is xN and of gas 2 is $(1-x)N$, where N is the total number of molecules.

If the tap on the pipe connecting the two vessels is opened, the gases will spontaneously mix, resulting in an increase in entropy, known as the **entropy of mixing**. As for the Joule expansion, we can imagine going from the starting state (gas 1 in the first vessel, gas 2 in the second vessel) to the final state (a homogeneous mixture of gas 1 and gas 2 distributed throughout both vessels) via a reversible route, so that we *imagine* a reversible expansion of gas 1 from xV into the combined volume V and a reversible expansion of gas 2 from $(1-x)V$ into the combined volume V. For an isothermal expansion of an ideal gas, the internal energy doesn't change and hence $T\,\mathrm{d}S = p\,\mathrm{d}V$ so that $\mathrm{d}S = (p/T)\,\mathrm{d}V = Nk_{\mathrm{B}}\,\mathrm{d}V/V$, using the ideal gas law. This means that the entropy of mixing for our problem is

$$\Delta S = xNk_{\mathrm{B}}\int_{xV}^{V}\frac{\mathrm{d}V_1}{V_1} + (1-x)Nk_{\mathrm{B}}\int_{(1-x)V}^{V}\frac{\mathrm{d}V_2}{V_2} \tag{14.39}$$

and hence

$$\Delta S = -Nk_{\mathrm{B}}(x\ln x + (1-x)\ln(1-x)). \tag{14.40}$$

This equation is plotted in Fig. 14.7. As expected, there is no entropy increase when $x = 0$ or $x = 1$. The maximum entropy change occurs when $x = \frac{1}{2}$ in which case $\Delta S = Nk_{\mathrm{B}}\ln 2$. This of course corresponds to the equilibrium state in which no further increase of entropy is possible.

This expression for $x = \frac{1}{2}$ also admits to a very simple statistical interpretation. Before the mixing of the gases takes place, we know that gas 1 is only in the first vessel and gas 2 is only in the second vessel. After mixing, each molecule can exist in two additional 'microstates' than before; for every microstate with a molecule of gas 1 on the left there is now an additional one with a molecule of gas 1 now on the right. Therefore Ω must be multiplied by 2^N and hence S must increase by $k_{\mathrm{B}}\ln 2^N$ which is $Nk_{\mathrm{B}}\ln 2$.

This treatment has an important consequence: distinguishability is an important concept! We have assumed that there is some tangible *difference* between gas 1 and gas 2, so that there is some way to label whether a particular molecule is gas 1 or gas 2. For example, if the two gases were nitrogen and oxygen, one could measure the mass of the molecules to determine which was which. But what if the two gases were actually the same? Physically, we would expect that mixing them would have no observable consequences, so there should be no increase in entropy. Thus mixing should only increase entropy if the gases really are distinguishable. We will return to this issue of distinguishability in Chapter 29.

14.7 Maxwell's demon

In 1867, James Clerk Maxwell came up with an intriguing puzzle via a thought experiment. This has turned out to be much more illuminating and hard to solve than he might ever have imagined. The thought experiment can be stated as follows: imagine performing a Joule expansion on a gas. A gas is initially in one chamber, which is connected via a closed tap to a second chamber containing only a vacuum (see Fig. 14.4). The tap is opened and the gas in the first chamber expands to fill both chambers. Equilibrium is established and the pressure in each chamber is now half of what it was in the first chamber at the start. The Joule expansion is formally irreversible as there is no way to get the gas back into the initial chamber without doing work. Or is there? Maxwell imagined that the tap was operated by a microscopic intelligent creature, now called **Maxwell's demon**, who was able to watch the individual molecules bouncing around close to the tap (see Fig. 14.8). If the demon sees a gas molecule heading from the second chamber back into the first, it quickly opens the tap and then shuts it straight away, just letting the molecule through. If it spots a gas molecule heading from the first chamber back into the second chamber, it keeps the tap closed. The demon does no work[5] and yet it can make sure that the gas molecules in the second chamber all go back into the first chamber. Thus it creates a pressure difference between the two chambers where none existed before the demon started its mischief.

[5] It does no work in the $p\mathrm{d}V$ sense, though it does do some in the brain sense.

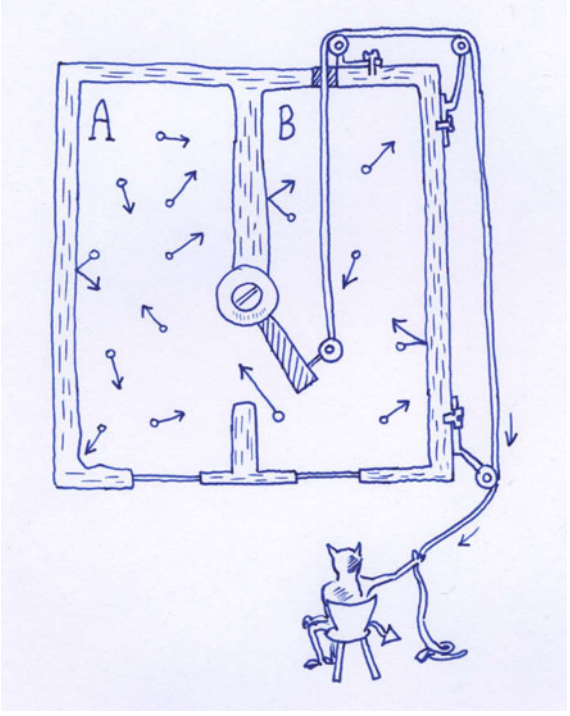


Fig. 14.8 Maxwell's demon watches the gas molecules in chambers A and B and *intelligently* opens and shuts the trap door connecting the chambers. The demon is therefore able to reverse the Joule expansion and only let molecules travel from B to A, thus *apparently* contravening the second law of thermodynamics.

Now, a similar demon could be employed to make hot molecules go the wrong way (i.e. so that heat flows the wrong way, from cold to hot – this in fact was Maxwell's original implementation of the demon), or even to sort out molecules of different types (and thus subvert the 'entropy of mixing', see Section 14.6). It looks as if the demon could therefore cause entropy to decrease in a system with no consequent increase in entropy anywhere else. In short, Maxwell's demon appears to make a mockery out of the second law of thermodynamics. How on earth does it get away with it?

Many very good minds have addressed this problem. One early idea was that the demon needs to make measurements of where all the gas molecules are, and to do this would need to shine light on the molecules; thus the process of observation of the molecules might be thought to rescue us from Maxwell's demon. However, this idea turned out not to be correct as it was found to be possible, even in principle, to detect a molecule with arbitrarily little work and dissipation. Remarkably, it turns out that because a demon needs to have a memory to operate (so that it can remember where it has observed a molecule and any other results of its measurement process), this act of storing *information* (actually it is the act of erasing information, as we will discuss below) is associated with an increase of entropy, and this increase cancels out any decrease in entropy that the demon might be able to effect in the system. This connection between information and entropy is an extremely important insight and will be explored in Chapter 15.

The demon is in fact a type of computational device that processes and stores information about the world. It is possible to design a computational process which proceeds entirely reversibly, and therefore has no increase in entropy associated with it. However, the act of erasing information is irreversible (as anyone who has ever failed to backup their data and then had their computer crash will testify). Erasing information always has an associated increase in entropy (of $k_\mathrm{B} \ln 2$ per bit, as we shall see in Chapter 15); Maxwell's demon can operate reversibly therefore, but only if it has a large enough hard disk that it doesn't ever need to clear space to continue operating. The Maxwell demon therefore beautifully illustrates the connection between entropy and information.

14.8 Entropy and probability

The entropy that you measure is due to the number of different states in which the system can exist, according to $S = k_\mathrm{B} \ln \Omega$ (eqn 14.36). However, each state may consist of a large number of microstates that we can't directly measure. Since the system could exist in any one of those microstates, there is extra entropy associated with them. An example should make this idea clear.

Example 14.5

A system has 5 possible equally likely states in which it can exist, and which of those states it occupies can be distinguished by some easy physical measurement. The entropy is therefore, using eqn 14.36,

$$S = k_{\mathrm{B}} \ln 5. \tag{14.41}$$

However, each of those 5 states is made up of 3 equally likely microstates and it is not possible to measure easily which of those microstates it is in. The extra entropy associated with these microstates is $S_{\mathrm{micro}} = k_{\mathrm{B}} \ln 3$. The system therefore *really* has $3 \times 5 = 15$ states and the total entropy is therefore $S_{\mathrm{tot}} = k_{\mathrm{B}} \ln 15$. This can be decomposed into

$$S_{\mathrm{tot}} = S + S_{\mathrm{micro}}. \tag{14.42}$$

Now let us suppose that a system can have N different, equally–likely microstates. As usual, it is hard to measure the details of these microstates directly, but let us assume that they are there. These microstates are divided into various groups (we will call these groups *macrostates*) with n_i microstates contained in the ith macrostate. The macrostates are easier to distinguish using experiment because they correspond to some macroscopic, measurable property. We must have that the sum of all the microstates in each macrostate is equal to the total number of microstates, so that

$$\sum_i n_i = N. \tag{14.43}$$

The probability P_i of finding the system in the ith macrostate is then given by

$$P_i = \frac{n_i}{N}. \tag{14.44}$$

Equation 14.43 then implies that $\sum P_i = 1$ as required. The total entropy is of course $S_{\mathrm{tot}} = k_{\mathrm{B}} \ln N$, though we can't measure that directly (having no information about the microstates which is easily accessible). Nevertheless, S_{tot} is equal to the sum of the entropy associated with the freedom of being able to be in different macrostates, which is our measured entropy S, and the entropy S_{micro} associated with it being able to be in different microstates within a macrostate. Putting this statement in an equation, we have

$$S_{\mathrm{tot}} = S + S_{\mathrm{micro}}, \tag{14.45}$$

which is identical to eqn 14.42. The entropy associated with being able to be in different microstates (the aspect we can't measure) is given by

$$S_{\mathrm{micro}} = \langle S_i \rangle = \sum_i P_i S_i, \tag{14.46}$$

where $S_i = k_\mathrm{B} \ln n_i$ is the entropy of the microstates in the ith macrostate and, to recap, P_i is the probability of a particular macrostate being occupied. Hence

$$\begin{aligned} S &= S_\mathrm{tot} - S_\mathrm{micro} \\ &= k_\mathrm{B}\left(\ln N - \sum_i P_i \ln n_i\right) \\ &= k_\mathrm{B} \sum_i P_i (\ln N - \ln n_i), \end{aligned} \tag{14.47}$$

and using $\ln N - \ln n_i = -\ln(n_i/N) = -\ln P_i$ (from eqn 14.44) yields **Gibbs' expression for the entropy**:

$$\boxed{S = -k_\mathrm{B} \textstyle\sum_i P_i \ln P_i.} \tag{14.48}$$

Example 14.6

Find the entropy for a system with Ω macrostates, each with probability $P_i = 1/\Omega$ (i.e. assuming the microcanonical ensemble).
Solution:
Using eqn 14.48, substitution of $P_i = 1/\Omega$ yields

$$S = -k_\mathrm{B} \sum_i P_i \ln P_i = -k_\mathrm{B} \sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = -k_\mathrm{B} \ln \frac{1}{\Omega} = k_\mathrm{B} \ln \Omega, \tag{14.49}$$

which is the same as eqn 14.36.

A connection between the Boltzmann probability and the expression for entropy in eqn 14.48 is demonstrated in the following example.

Example 14.7

Maximise $S = -k_\mathrm{B} \sum_i P_i \ln P_i$ (eqn 14.48) subject to the constraints that $\sum P_i = 1$ and $\sum_i P_i E_i = U$.

Solution:

[6] See Appendix C.13.

Use the method of Lagrange multipliers,[6] in which we maximize

$$\frac{S}{k_\mathrm{B}} - \alpha \times (\text{constraint 1}) - \beta \times (\text{constraint 2}) \tag{14.50}$$

where α and β are Lagrange multipliers. Thus we vary this expression with respect to one of the probabilities P_j and get

$$\frac{\partial}{\partial P_j}\left(\sum_i -P_i \ln P_i - \alpha P_i - \beta P_i E_i\right) = 0, \tag{14.51}$$

so that

$$-\ln P_j - 1 - \alpha - \beta E_j = 0. \tag{14.52}$$

This can be rearranged to give

$$P_j = \frac{\mathrm{e}^{-\beta E_j}}{\mathrm{e}^{1+\alpha}}, \tag{14.53}$$

so that with $Z = \mathrm{e}^{1+\alpha}$ we have

$$P_j = \frac{\mathrm{e}^{-\beta E_j}}{Z} \tag{14.54}$$

which is our familiar expression for the Boltzmann probability (eqn 4.13).

Chapter summary

- Entropy is defined by $\mathrm{d}S = đQ_{\mathrm{rev}}/T$.
- The entropy of an isolated system tends to a maximum.
- The entropy of an isolated attains this maximum at equilibrium.
- The laws of thermodynamics can be stated as follows:
  (1) $U_{\mathrm{Universe}} =$ constant.
  (2) S_{Universe} can only increase.
- These can be combined to give $\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$ which always holds.
- The statistical definition of entropy is $S = k_{\mathrm{B}} \ln \Omega$.
- The general definition of entropy, due to Gibbs, is $S = -k_{\mathrm{B}} \sum_i P_i \ln P_i$.

Exercises

(14.1) A mug of tea has been left to cool from 90°C to 18°C. If there is 0.2 kg of tea in the mug, and the tea has specific heat capacity 4200 J K^{-1} kg^{-1}, show that the entropy of the tea has decreased by 185.7 J K^{-1}. Comment on the sign of this result.

(14.2) In a free expansion of a perfect gas (also called Joule expansion), we know U does not change, and no work is done. However, the entropy must increase because the process is irreversible. Are these statements compatible with the first law $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$?

(14.3) A 10 Ω resistor is held at a temperature of 300 K. A current of 5 A is passed through the resistor for 2 minutes. Ignoring changes in the source of the current, what is the change of entropy in (a) the resistor and (b) the Universe?

(14.4) Calculate the change of entropy
(a) of a bath containing water, initially at 20°C, when it is placed in thermal contact with a very

large heat reservoir at 80°C,
(b) of the reservoir when process (a) occurs,
(c) of the bath and of the reservoir if the bath is brought to 80°C through the operation of a Carnot engine between them.
The bath and its contents have total heat capacity $10^4\,\mathrm{J\,K^{-1}}$.

(14.5) A block of lead of heat capacity 1 kJ K^{-1} is cooled from 200 K to 100 K in two ways.
(a) It is plunged into a large liquid bath at 100 K.
(b) The block is first cooled to 150 K in one liquid bath and then to 100 K in another bath.
Calculate the entropy changes in the system comprising block plus baths in cooling from 200 K to 100 K in these two cases. Prove that in the limit of an infinite number of intermediate baths the total entropy change is zero.

(14.6) Calculate the changes in entropy of the Universe as a result of the following processes:
(a) A capacitor of capacitance 1 μF is connected to a battery of e.m.f. 100 V at 0°C. (NB think carefully about what happens when a capacitor is charged from a battery.)
(b) The same capacitor, after being charged to 100 V, is discharged through a resistor at 0°C.
(c) One mole of gas at 0°C is expanded reversibly and isothermally to twice its initial volume.
(d) One mole of gas at 0°C is expanded reversibly and adiabatically to twice its initial volume.
(e) The same expansion as in (f) is carried out by opening a valve to an evacuated container of equal volume.

(14.7) Consider n moles of a gas, initially confined within a volume V and held at temperature T. The gas is expanded to a total volume αV, where α is a constant, by (a) a reversible isothermal expansion and (b) removing a partition and allowing a free expansion into the vacuum. Both cases are illustrated in Fig. 14.9. Assuming the gas is ideal, derive an expression for the change of entropy of the gas in each case.

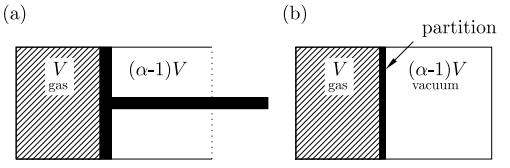


Fig. 14.9 Diagram showing n moles of gas, initially confined within a volume V.

Repeat this calculation for case (a), assuming that the gas obeys the van der Waals equation of state

$$\left(p + \frac{n^2 a}{V^2}\right)(V - nb) = nRT. \tag{14.55}$$

Show further that for case (b) the temperature of the van der Waals gas falls by an amount proportional to $(\alpha - 1)/\alpha$.

(14.8) The probability of a system being in the ith microstate is

$$P_i = \mathrm{e}^{-\beta E_i}/Z, \tag{14.56}$$

where E_i is the energy of the ith microstate and β and Z are constants. Show that the entropy is given by

$$S/k_\mathrm{B} = \ln Z + \beta U, \tag{14.57}$$

where $U = \sum_i P_i E_i$ is the internal energy.

(14.9) Use the Gibbs expression for entropy (eqn 14.48) to derive the formula for the entropy of mixing (eqn 14.40).

Julius Robert Mayer (1814-1878)

Robert Mayer studied medicine in Tübingen and took the somewhat unusual career route of signing up as a ship's doctor with a Dutch vessel bound for the East Indies. While letting blood from sailors in the tropics, he noticed that their venous blood was redder than observed back home and concluded that the metabolic oxidation rate in hotter climates was slower. Since a constant body temperature was required for life, the body must reduce its oxidation rate because oxidation of material from food produces internal heat. Though there was some questionable physiological reasoning in his logic, Mayer was on to something. He had realised that energy was something that needed to be *conserved* in any physical process. Back in Heilbronn, Germany, Mayer set to work on a measurement of the mechanical equivalent of heat and wrote a paper in 1841 which was the first statement of the conservation of energy (though he used the word 'force'). Mayer's work predated the ideas of Joule and Helmholtz (though his experiment was not as accurate as Joule's) and his notion of the conservation of energy had a wider scope than that of Helmholtz; not only were mechanical energy and heat convertible, but his principle could be applied to tides, meteorites, solar energy and living things. His paper was eventually published in 1842, but received little acclaim. A later more detailed paper in 1845 was rejected and he published it privately.



Fig. 14.10 Robert Mayer

Mayer then went through a bit of a bad patch, to put it mildly: others began to get the credit for ideas he thought he had pioneered, three of his children died in the late 1840's and he attempted suicide in 1850, jumping out of a third-story window, but only succeeding in permanently laming himself. In 1851 he checked into a mental institution where he received sometimes brutal treatment and was discharged in 1853, with the doctors unable to offer him any hope of a cure. In 1858, he was even referred to as being dead in a lecture by Liebig (famous for his condenser, and editor of the journal that had accepted Mayer's 1842 paper). Mayer's scientific reputation began to recover in the 1860's and he was awarded the Copley Medal of the Royal Society of London in 1871, the year after they awarded it to Joule.

James Prescott Joule (1818-1889)

James Joule was the son of a wealthy brewer in Salford, near Manchester, England. Joule was educated at home, and his tutors included John Dalton, the father of modern atomic theory. In 1833, illness forced his father to retire, and Joule was left in charge of the family brewery. He had a passion for scientific research and set up a laboratory, working there in the early morning and late evening so that he could continue his day job. In 1840, he showed that the heat dissipated by an electric current I in a resistor R was proportional to I^2R (what we now call **Joule heating**). In 1846, Joule discovered the phenomenon of magnetostriction (by which a magnet changes its length when magnetized). However Joule's work did not impress the Royal Society and he was dismissed as a mere provincial dilettante. However, Joule was undeterred and he decided to work on the convertibility of energy and to try to measure the mechanical equivalent of heat.



Fig. 14.11 James Joule

In his most famous experiment he measured the increase in temperature of a thermally insulated barrel of water, stirred by a paddle-wheel which was driven by a falling weight. But this was just one of an exhaustive series of meticulously performed experiments which aimed to determine the mechanical equivalent of heat, using electrical circuits, chemical reactions, viscous heating, mechanical contraptions and gas compression. He even attempted to measure the temperature difference between water at the top and bottom of a waterfall, an opportunity afforded to him by being in Switzerland on his honeymoon!

Joule's obsessive industry paid off: his completely different experimental methods gave consistent results.

Part of Joule's success was in designing thermometers with unprecedented accuracy which could measure temperature changes as small as 1/200 degrees Fahrenheit. This was necessary as the effects he was looking for tended to be small. His methods proved to be accurate and even his early measurements were within several percent of the modern accepted value of the mechanical equivalent of heat, and his 1850 experiment was within 1 percent. However, the smallness of the effect led to scepticism, particularly from the scientific establishment, who had all had proper educations, didn't spend their days making beer and knew that you couldn't measure temperature differences as tiny as Joule claimed to have observed.

However the tide began to turn in Joule's favour in the late 1840's. Helmholtz recognized Joule's contribution to the conservation of energy in his paper of 1847. In the same year, Joule gave a talk at a British Association meeting in Oxford where Stokes, Faraday and Thomson were in attendance. Thomson was intrigued and the two struck up a correspondence, resulting in a fruitful collaboration between the two between 1852 and 1856. They measured the temperature fall in the expansion of a gas, and discovered the Joule–Thomson effect.

Joule refused all academic appointments, preferring to work independently. Though without advanced education, Joule had excellent instincts and was an early defender of the kinetic theory of gases, and felt his way towards a kinetic theory of heat, perhaps because of his youthful exposure to Dalton's teachings. On Joule's gravestone is inscribed the number '772.55', the number of foot-pounds required to heat a pound of water by one degree Fahrenheit. It is fitting that today, mechanical and thermal energy are measured in the same unit: the Joule.

Rudolf Clausius (1822-1888)

Rudolf Clausius studied mathematics and physics in Berlin, and did his doctorate in Halle University on the colour of the sky. Clausius turned his attention to the theory of heat and, in 1850, he published a paper which essentially saw him picking up the baton left by Sadi Carnot (via an 1834 paper by Emile Clapeyron) and running with it. He defined the internal energy, U, of a system and wrote that the change of heat was given by $\mathrm{d}Q = \mathrm{d}U + (1/J)p\,\mathrm{d}V$, where the factor J (the mechanical equivalent of heat) was necessary to convert mechanical energy $p\,\mathrm{d}V$ into the same units as thermal energy (a conversion which in today's units is, of course, unnecessary). He also showed that in a Carnot process, the integral round a closed loop of $f(T)\,\mathrm{d}Q$ was zero, where $f(T)$ was some function of temperature.



Fig. 14.12 Rudolf Clausius

His work brought him a professorship in Berlin, though he subsequently moved to chairs in Zürich (1855), Würzburg (1867) and Bonn (1869). In 1854, he wrote a paper in which he stated that heat cannot of itself pass from a colder to a warmer body, a statement of the second law of thermodynamics. He also showed that his function $f(T)$ could be written (in modern notation) as $f(T) = 1/T$. In 1865 he was ready to give $f(T)\,\mathrm{d}Q$ a name, defining the entropy (a word he made up to sound like 'energy' but contain 'trope' meaning 'turning', as in the word 'heliotrope', a plant which turns towards the Sun) using $\mathrm{d}S = \mathrm{d}Q/T$ for a reversible process. He also summarized the first and second laws of thermodynamics by stating that the energy of the world is constant and its entropy tends to a maximum.

When Bismarck started the Franco-German war, Clausius patriotically ran a volunteer ambulance corps of Bonn students in 1870–1871, carrying off the wounded from battles in Vionville and Gravelotte. He was wounded in the knee, but received the Iron Cross for his efforts in 1871. He was no less zealous in defending Germany's preeminence in thermal physics in various priority disputes, being provoked into siding with Mayer's claim over Joule's, and in various debates with Tait, Thomson and Maxwell. Clausius however showed little interest in the work of Boltzmann and Gibbs that aimed to understand the molecular origin of the irreversibility that he had discovered and named.

Information theory

15

In this chapter we are going to examine the concept of information and relate it to thermodynamic entropy. At first sight, this seems a slightly crazy thing to do. What on earth do something to do with heat engines and something to do with bits and bytes have in common? It turns out that there is a very deep connection between these two concepts. To understand why, we begin our account by trying to formulate one definition of information.

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15.1 Information and Shannon entropy

Consider the following three true statements about Isaac Newton (1643–1727) and his birthday.[1]

(1) Isaac Newton's birthday falls on a particular day of the year.

(2) Isaac Newton's birthday falls in the second half of the year

(3) Isaac Newton's birthday falls on the 25th of a month.

[1] The statements take as prior information that Newton was born in 1643 and that the dates are expressed according to the calendar which was used in his day. The Gregorian calendar was not adopted in England until 1742.

The first statement has, by any sensible measure, no information content. *All* birthdays fall on a particular day of the year. The second statement has more information content: at least we now know which half of the year his birthday is. The third statement is much more specific and has the greatest information content.

How do we quantify information content? Well, one property we could notice is that the greater the probability of the statement being true *in the absence of any prior information*, the less the information content of the statement. Thus if you knew no prior information about Newton's birthday, then you would say that statement 1 has probability $P_1 = 1$, statement 2 has probability $P_2 = \frac{1}{2}$, and statement 3 has probability[2] $P_3 = \frac{12}{365}$; so as the probability decreases, the information content increases. Moreover, since the useful statements 2 and 3 are independent, then if you are given statements 2 and 3 together, their information contents should *add*. Moreover, the probability of statements 2 and 3 *both* being true, in the absence of prior information, is $P_2 \times P_3 = \frac{6}{365}$. Since the probability of two independent statements being true is the *product* of their individual probabilities, and since it is natural to assume that information content is *additive*, one is motivated to adopt the definition of information which was proposed by Claude Shannon (1916–2001) as follows:

[2] We are using the fact that 1643 was not a leap year!

The **information** content Q of a statement is defined by

$$Q = -k \log P, \tag{15.1}$$

where P is the probability of the statement and k is a positive constant.[3] If we use $\log_2$ (log to the base 2) for the logarithm in this expression and also $k = 1$, then the information Q is measured in **bits**. If instead we use $\ln \equiv \log_e$ and choose $k = k_B$, then we have a definition which, as we shall see, will match what we have found in thermodynamics. In this chapter, we will stick with the former convention since bits are a useful quantity with which to think about information.

[3] We need k to be a positive constant so that as P goes up, Q goes down.

Thus, if we have a set of statements with probability P_i, with corresponding information $Q_i = -k \log P_i$, then the average information content S is given by

$$S = \langle Q \rangle = \sum_i Q_i P_i = -k \sum_i P_i \log P_i. \tag{15.2}$$

The average information is called the **Shannon entropy**.

Example 15.1

- A fair die produces outcomes 1, 2, 3, 4, 5 and 6 with probabilities $\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$. The information associated with each outcome is $Q = -k \log \frac{1}{6} = k \log 6$ and the average information content is then $S = k \log 6$. Taking $k = 1$ and using log to the base 2 gives a Shannon entropy of 2.58 bits.
- A biased die produces outcomes 1, 2, 3, 4, 5 and 6 with probabilities $\frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{2}$. The information contents associated with the outcomes are $k \log 10$,$k \log 10$, $k \log 10$, $k \log 10$, $k \log 10$ and $k \log 2$. (These are 3.32, 3.32, 3.32, 3.32, 3.32 and 1 bit respectively.) If we take $k = 1$ again, the Shannon entropy is then $S = k(5 \times \frac{1}{10} \log 10 + \frac{1}{2} \log 2) = k(\log \sqrt{20})$ (this is 2.16 bits). This Shannon entropy is smaller than in the case of the fair die.

The Shannon entropy quantifies how much information we gain, on average, following a measurement of a particular quantity. (Another way of looking at it is to say the Shannon entropy quantifies the amount of *uncertainty* we have about a quantity *before* we measure it.) To make these ideas more concrete, let us study a simple example in which there are only two possible outcomes of a particular random process (such as the tossing of a coin, or asking the question 'will it rain tomorrow?').

Example 15.2

What is the Shannon entropy for a Bernoulli[4] trial (a two-outcome random variable) with probabilities P and $1-P$ of the two outcomes?
Solution:

$$S = -\sum_i P_i \log P_i = -P \log P - (1-P)\log(1-P), \tag{15.3}$$

where we have set $k = 1$. This behaviour is sketched in Fig. 15.1. The Shannon entropy has a maximum when $p = \frac{1}{2}$ (greatest uncertainty about the outcome, or greatest information gained, 1 bit, following a trial) and a minimum when $p = 0$ or 1 (least uncertainty about the outcome, or least information gained, 0 bit, following a trial).

The information associated with each of the two possible outcomes is also shown in Fig. 15.1 as dashed lines. The information associated with the outcome having probability P is given by $Q_1 = -\log_2 P$ and decreases as P increases. Clearly when this outcome is very unlikely (P small) the information associated with getting that outcome is very large (Q_1 is many bits of information). However, such an outcome doesn't happen very often so it doesn't contribute much to the average information (i.e. to the Shannon entropy, the solid line in Fig. 15.1). When this outcome is almost certain (P almost 1) it contributes a lot to the average information but has very little information content. For the other outcome, with probability $1-P$, $Q_2 = -\log_2(1-P)$ and the behaviour is simply a mirror image of this. The maximum average information is when $P = 1-P = \frac{1}{2}$ and both outcomes have 1 bit of information associated with them.

[4] James Bernoulli (1654–1705).

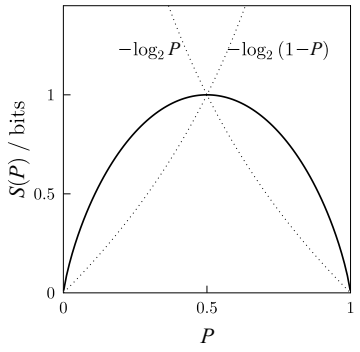


Fig. 15.1 The Shannon entropy of a Bernoulli trial (a two-outcome random variable) with probabilities P and $1-P$ of the two outcomes. The units are chosen so that the Shannon entropy is in bits. Also shown is the information associated with each outcome (dashed lines).

15.2 Information and thermodynamics

Remarkably, the formula for Shannon entropy in eqn 15.2 is identical (apart from whether you take your constant as k or k_B) to Gibbs' expression for thermodynamic entropy in eqn 14.48. This gives us a useful perspective on what thermodynamic entropy is. It is a measure of our uncertainty of a system, based upon our limited knowledge of its properties and ignorance about which of its microstates it is in. In making inferences on the basis of partial information, we can assign probabilities on the basis that we maximize entropy subject to the constraints provided by what is known about the system. This is exactly what we did in Example 14.7, when we maximized the Gibbs entropy of an isolated system subject to the constraint that the total energy U was constant; hey presto, we found that we recovered the Boltzmann probability distribution. With this viewpoint, one can begin to understand thermodynamics from an information theory viewpoint.